

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: 2.5 RATES OF REACTIONS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the Disappearing Cross. Mwalimu Achieng's class in Nakuru County has been asking: Why do some chemical reactions happen in a flash while others take days, and how can we control the speed of reactions to make life in Kenya safer, healthier and more productive? Use your science notebook, the class data tables and your Driving Question Board (DQB) to help you. Write your answers in the space under each prompt. Use full sentences, include numbers with units where you can, and use the words particles, effective collisions and activation energy. Remember the claim, evidence and reasoning structure. In Section 3 you will use a table of results.

Section 1: Explain the Disappearing Cross (Claim, Evidence, Reasoning)

In the anchor demonstration, 10 cm³ of 1.0 mol/dm³ hydrochloric acid was put in each of two beakers and 50 cm³ of 0.15 mol/dm³ sodium thiosulfate was added. Beaker A was at 10 °C and Beaker B was at 50 °C. The black cross disappeared at 35 s in one beaker and at 270 s in the other.

- (a) Write a CLAIM that answers Wanjiru's question: why did one beaker reach the end point faster than the other?
- (b) Give EVIDENCE from the demo: the times, how many times faster, and the rate for each beaker using $\text{rate} = 1/t$.
- (c) Write your REASONING using particles, collisions, kinetic energy and activation energy.
- (d) State what was the same in both beakers, and what you can say about the amount of sulfur made at the end.

Write your answer here:

Section 2: Fair Testing and Measuring Rate

Amina wants to test whether temperature changes the rate, using the cross method.

(a) Name the independent variable, the dependent variable and at least three control variables for her test.

(b) Amina says: 'I will use a different observer and a bigger beaker for the hot test, because it is faster anyway.' Explain why this plan is not a fair test.

(c) Explain why a shorter time for the cross to disappear gives a bigger rate.

(d) Name two other ways, besides the time for a precipitate to hide a cross, that the rate of a reaction can be measured.

Write your answer here:

Section 3: Using Evidence from the Tables

Here is the class-mean data from Lessons 3 and 4. All used 10 cm³ of 1.0 mol/dm³ HCl and a total thiosulfate volume of 50 cm³.

Temperature test (0.15 mol/dm³ thiosulfate):

Temperature (C)	Time for cross to disappear (s)	Rate (x10 ⁻³ s ⁻¹)
10	270	3.7
25	110	9.1
40	52	19.2
55	26	38.5

Concentration test (all at 25 C):

Sodium thiosulfate (mol/dm ³)	Time (s)	Rate (x10 ⁻³ s ⁻¹)
0.15	110	9.1

0.12	138	7.2
0.09	183	5.5
0.06	275	3.6

- (a) Describe the pattern in rate as temperature rises from 10 C to 55 C.
 (b) The anchor demo at 50 C took 35 s. Show whether this fits your pattern in (a).
 (c) Describe the pattern between concentration and rate, and use the numbers to say how rate and concentration are related.
 (d) Explain the concentration pattern using particles. Why does the energy of the particles not explain it?

Write your answer here:

Section 4: Surface Area, Catalysts and Light

- (a) In Lesson 5, 0.5 g of marble chips (CaCO_3) in 25 cm³ of 1.0 mol/dm³ HCl at 25 C gave initial rates over the first 30 s of 0.40 cm³/s for large chips, 0.93 cm³/s for small chips and 2.07 cm³/s for powder. Explain why powder gave the fastest rate, and say what happened to the final volume of CO₂ for all three.
 (b) In Lesson 6, 10 cm³ of 3% hydrogen peroxide with detergent gave 0 cm of foam with no additive, 0 cm with sand, 0.5 cm with a boiled liver cube and 9 cm with a fresh liver cube after 30 s. Explain why fresh liver made so much foam, using the words catalyst, enzyme and activation energy. Why did the boiled liver not work well?
 (c) Kiprono says light is a catalyst. Do you agree? Give a reason and an example.

Write your answer here:

Section 5: Solve a Kenyan Problem and Reflect on Your Model

A trader in Kisumu finds that her fresh fish from Lake Victoria spoils within hours. A farmer in Kericho wants to speed up composting of tea waste.

- (a) For the fish trader, choose one way to slow down spoilage reactions. Explain, using particles and collisions, why it works.
- (b) For the Kericho farmer, choose one factor to change to speed up composting. Say how you would change it and explain why it works.
- (c) Use the idea of an optimum to explain why the enzyme amylase in your body works best near 37 C and pH 7, and what you would expect if conditions were far from this.
- (d) Describe how your particle model has changed since Lesson 1, and state one DQB question that is now answered.
- (e) Write one sentence of advice for your community about controlling reaction rates, and name a career that uses this knowledge.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim, evidence and reasoning for the anchor phenomenon	Gives a clear claim, uses exact data (35 s, 270 s, 7.7 times, rates 28.6 and $3.7 \times 10^{-3} \text{ s}^{-1}$) and links more kinetic energy, more frequent and energetic collisions, and activation energy to more effective collisions per second.	Gives a claim and uses most of the data, with reasoning that mentions collisions and energy but may leave out activation energy or effective collisions.	Claim is vague or reasoning says only that particles move faster or that the reaction is hotter; little or no data is used.
Fair testing and use of rate = $1/t$	Names independent, dependent and control variables correctly, explains why changing several things is unfair, and correctly explains rate = $1/t$.	Names most variables and gives a basic reason for fairness; rate = $1/t$ is used correctly with small gaps in the explanation.	Confuses variables or does not show why the plan is unfair; rate and time are mixed up.
Interpreting data tables	Describes trends accurately with numbers, checks that 35 s fits between 52 s and 26 s, and shows rate roughly proportional to concentration (about 60).	Describes trends correctly but with few numbers or without testing proportionality or the 35 s link.	Misreads the tables or states trends without evidence.
Understanding of surface area, catalysts and light	Explains surface area, catalysts (lower activation energy, not used up, enzyme) and light correctly, and states that factors change speed, not the final amount.	Explains most factors correctly with small gaps, such as missing activation energy or the final amount.	Mixes up catalysts and light or gives no particle explanation.

Applying the model to Kenyan contexts and reflecting	Chooses suitable factors for the fish and compost problems, explains them with collision theory, links to optimum conditions for enzymes, and clearly compares the model with Lesson 1.	Suggests suitable actions with partial particle explanations and a general reflection.	Gives actions without reasons, or no reflection on how the model changed.
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